

JANNET AKANKSHA EKKA

Applied AI Engineer • LLM Agents • AI Evaluation • Agent Security

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PROFESSIONAL SUMMARY

I build LLM agents and the tests that show whether they work. I spent four years leading QA automation at Deloitte, then finished Rank 1 in an AI/ML postgrad at UT Austin McCombs. Since 2024 I have built a patent-pending AI trading agent on my own, directing AI coding agents and reviewing every merge myself. The lesson I trust most from it: measure an agent before you believe it. When I graded my six trading agents on independent data, all six called direction at a coin flip, so I rebuilt the system around that result.

TECHNICAL SKILLS

AI Evaluation	Eval harnesses (recall@k, MRR, paired bootstrap), counterfactual faithfulness tests, adversarial agent testing by tool-call interception, overfitting gates (CPCV, Deflated Sharpe, PBO, FDR), replay through production code, liveness checks on production logs, pytest, Selenium, test strategy
GenAI & Agents	Google ADK, Model Context Protocol (servers written from scratch), A2A SDK, Gemini (Vertex AI and AI Studio), Imagen, OpenServ SDK, multi-agent orchestration, prompt engineering, structured output, RAG (chunking, BM25, reciprocal-rank fusion, cross-encoder reranking), Hugging Face
AI-Native Delivery	Claude Code, spec-driven development, agent-directed implementation, pull-request review and merge control, production log triage, defect isolation
ML & Quant	scikit-learn, TensorFlow, Keras, PyTorch, CatBoost, Optuna (TPE), contextual bandits, Kelly sizing, time-series, NLP, computer vision (ResNet, Fast R-CNN), NVIDIA cuDF / RAPIDS
Cloud & Infra	Oracle Cloud, Cloudflare Workers; GCP: Compute Engine, Cloud Run, Cloud Functions, BigQuery (+ vector search), AlloyDB & pgvector, Firestore, Secret Manager, Vertex AI; AWS: S3, EC2, RDS, SageMaker, Texttract, Comprehend; Docker, systemd, GitHub Actions
Engineering	Python and SQL are the languages I take into a coding interview. TypeScript and JavaScript I have shipped and read fluently, and I would be ramping in them rather than leading. FastAPI, Flask, Streamlit, Django, React, Next.js, Postgres, REST and WebSocket APIs

FLAGSHIP PROJECT

Smart Money Trading (SMT) — Explainable AI Trading Agent • Founder & sole engineer
Patent pending: Indian provisional application No. 202631090789, filed myself in July 2026
Live: smt-weex-trading-bot.jannet-ekka.workers.dev • Public layer: github.com/JannetEkka/smt-apac • WEEX feature

Jan 2024 – Present

- Built it alone by specifying the work and directing AI coding agents; nothing merges without my review. **91,000 lines of Python, 3,000+ tests, 344 merged pull requests**, and a register of **611 measured findings** that decides what ships.
- Graded my own multi-agent committee (six persona agents and a learned Judge) on independent market data. **All six called direction at a coin flip.** Rebuilt on that result: written rules with replayed records open every trade, and the agents became advisors graded on every call, earning a veto only after 30 proven warnings.
- Every explanation passes a **counterfactual test** before a user sees it: flip the advisor's input and re-run the decision; if the decision doesn't change, the reason is dropped. The learning loop retunes only through a **CPCV + Deflated Sharpe + PBO + FDR gate**, which still holds a forecaster that scored AUC 0.723 on a BTC forward test in shadow.
- Tested **1,800 strategy variants, 36 million simulated trades** from 2020 to 2026 at six exchanges' published fees, replayed through the production code. Nothing holding a position under four hours cleared its fees. The crash-buy rule that did: 312 trades, 73% won, +4.76% average after fees, positive in every year.
- Audited 98 AI-written merges line by line and found two failures the green test suite missed: an order-format bug that had **rejected 1,137 live orders over three months**, and a memory freeze that kept the daemon down 60% of the time for ten days. A fix now counts as done only when the production log shows it.
- Runs 24/7 and places real orders on WEEX's exchange-granted simulated account (about \$10K). When the GCP billing account closed, I **moved the live system to a zero-cost stack in two days** (Oracle Cloud free-tier VM, Cloudflare Workers, Google AI Studio's free tier) and it re-adopted its three open positions from committed state. Any substituted default logs a warning and a liveness flag, a rule that has caught four multi-week data bugs.
- Built **SMT World**, the product for users: a Three.js/WebGL world on Cloudflare Workers where people watch the agent trade and ask it why, answered by Gemini from the live book.

Stack: Python, pandas, Optuna, scikit-learn, CatBoost, Gemini, Oracle Cloud, Cloudflare Workers, BigQuery, systemd, pytest, GitHub Actions, Three.js

SELECTED AI PROJECTS

Wiretrap — Adversarial Security Testing for AI Agents • Gen AI Academy APAC, Cohort 3
github.com/JannetEkka/wiretrap

2026

- A security tester that checks which tools an agent actually called. Most scanners grade the agent's reply. This one reads the tool-call log.
- Every tool the agent declares is cloned as a dead copy with the same name and JSON Schema that records the call and returns fake data. Attacks run in separate Gemini conversations and a judge model reads the log.
- Caught a Google codelab agent saying "I don't have access or authorization to modify configuration files" after it had already run `ls -la` through its shell tool. Text scanners pass that; Wiretrap flags it with the arguments.
- A run with nothing classified reports **INCONCLUSIVE**, never HARDENED. Approved fixes are re-run against the attacks that broke them, and the report says when one didn't hold. 8 attack categories, 9.4K lines of TypeScript.

github.com/JannetEkka/smartdesk • Eval write-up: evals/RESULTS.md

- A root **ADK orchestrator** routing to inbox, calendar and knowledge-base sub-agents on Gemini and Cloud Run. Gmail and Calendar are reached through **MCP servers I wrote** against the protocol SDK.
- Built the **RAG evaluation harness before touching retrieval**: 40 labelled questions over 120 notes, recall@k and MRR@k, paired bootstrap at 10,000 resamples. Baseline recall@5 **0.963**.
- **The harness killed the change it was built to justify**. A cross-encoder reranker won on the development embedder (MRR@10 +0.094, p=0.04), then fell to +0.028 with a CI spanning zero on the production embedder, at the cost of 190 MB of PyTorch and search going from 29 ms to 2,382 ms. The baseline shipped.

Smart Money Tracker — Multi-Agent Blockchain Analytics • Winner, Best DeFi Application

2024

- Multi-agent whale tracking across Ethereum, Polygon and BNB Chain on the OpenServ SDK, with Alchemy webhooks for sub-second monitoring and confidence-scored detection of accumulation, distribution and coordinated moves. SMT grew out of it.

Applied ML Portfolio • 13 end-to-end projects, UT Austin McCombs PGP

2024 – 2025

- ResNet50 vehicle classification with Fast R-CNN detection over 16,185 images and 196 classes; churn prediction at 81.5%; semiconductor yield from 591 sensor signals; financial-news sentiment with Mistral-7B; SQL analytics. All at jannetekka.github.io/DSProjects.

PROFESSIONAL EXPERIENCE

JTech Solutions (pre-incorporation) — Founder & AI Engineer

Jan 2024 – Present • Kolkata

- Building SMT and SMT World (above) as the company's first product, while a full-time family carer and alongside the PGP in AI/ML.

Deloitte Consulting LLP — Test Automation Engineer (Analyst)

Aug 2019 – Jan 2023 • Bangalore

- **Led a 6-member QA engineering team** on automated testing for Fortune 500 clients including AT&T Consumer Connect, Hewlett Packard Enterprise and The Dental Supply Company.
 - Built Selenium and Katalon Studio automation that **improved test execution efficiency by 83%**; Jenkins-based reporting cut manual analysis time by **75%**.
 - Wrote Python analysis and Tableau dashboards tracking quality across **343 business components**, cutting test failures by 25%; analysed **50,000+ SAP Hybris transactions** to isolate performance bottlenecks.
 - Joined as a QA automation intern in Jan 2019 (North Dakota Health and Human Services website, POEDI XML testing); converted to analyst after six months. 2 Spot Awards and an Applause Award.
- Tech:** Selenium, Katalon Studio, Python, JavaScript, Jenkins, Salesforce, SAP Hybris, Tableau, JIRA, Git

AutoKorrekt — Lead Frontend Developer (EdTech startup MVP)

May 2024 – Jan 2025

- Led the frontend of an AI answer-evaluation platform for teachers: PDF processing with coordinate-based extraction over **AWS Textract**, a bilingual English/German feedback interface on Amazon Comprehend, and the ML data flow across S3, RDS and SageMaker; scaled to **1,000+ concurrent submissions**.

EDUCATION

Post Graduate Program in AI & Machine Learning • UT Austin McCombs & Great Lakes

Mar 2025

Rank 1 in batch • GPA 4.09/5

Bachelor of Technology, Information Technology • KIIT University, Bhubaneswar

2015 – 2019

CGPA 7.17/10

ACHIEVEMENTS & CERTIFICATIONS

- **Provisional patent filed**, India, App. No. 202631090789 (Jul 2026), covering the SMT decision and verification architecture.
- **WEEX AI Wars I: #1 on the preliminary leaderboard** of 230+ teams, then one of 37 finalists in the live AI trading competition; featured in WEEX's coverage and two live AMAs.
- **Winner, Best DeFi Application**, OpenServ × Hack2skill hackathon; runner-up, Best Website Application.
- **Top 101 of 1,500+ teams**, Google Cloud Gen AI Academy APAC 2026 Cohort 1 hackathon (SmartDesk).
- **Google Cloud Gen AI Academy 2025** (ID 2025H2S04GENAI-A01917) and APAC 2026 Cohorts 1, 2 and 3; Gen AI Exchange Hackathon 2025; Agentic AI Day; Asha AI Hackathon 2025.
- **CPSAT**, Certified Professional Selenium Automation Testing, 2022. All credentials on [Credly](#) and [Google Skills](#).